

# Sushant S. Mahajan

Solar Astrophysicist • Data Scientist • AI for Science  
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## SUMMARY

Research scientist and computational physicist building data-driven systems for solar physics, scientific inference, and time-series analysis.

## EXPERIENCE

Research Scientist, W. W. Hansen Experimental Physics Laboratory, Stanford University 2024–present

- Lead research on solar dynamics and large-scale flows from HMI/SDO observations.
- Produce new metadata and analysis workflows for solar magnetogram and velocity-field data.
- Contribute to an agentic AI-powered JSOC data-commons effort for public query and visualization.

Postdoctoral Fellow, W. W. Hansen Experimental Physics Laboratory, Stanford University 2022–2024

- Published first-author work on differential rotation, torsional oscillation, and meridional flow.
- Helped deliver the Stanford Solar Observatories Group data-sharing effort recognized in 2026.
- Collaborated across research, software, and data management teams on mission-facing analysis.

Solar Physics Postdoctoral Fellow, Institute for Astronomy, University of Hawai'i 2019–2022

- Built analysis pipelines for active-region inflows, solar-cycle trends, and helioseismic flow inference.
- Contributed to forecasting-oriented work on rare events and solar flare data.

## SELECTED IMPACT

- NASA Silver Group Achievement Award for JSOC flood recovery efforts 2025
- Stanford University Libraries Data Sharing Prize, Stanford Solar Observatories Group 2026
- Editor's Choice, *Solar Physics*, for *The Sun's Large-Scale Flows I* 2024
- Prepared and published the SWAN-SF benchmark dataset used in solar-flare AI work 2020
- Presentation awards: Best Young Presenter Award (IAU Symposium 340), Best Young Scientist Poster Award (IAU Symposium 335), Honorary Mention (SPD meeting) 2018–2016

## SELECTED TALKS & MEDIA

“Unveiling the Sun: Exploring the Wonders of Solar Astrophysics,” KIPAC public lecture and “The Influence of Active Regions on Plasma Flows Inside the Sun,” UCAR/HAO colloquium 2023

“Unveiling the Solar Poles: Addressing Projection Artifacts and Revealing Circumpolar Features in HMI Magnetograms,” AAS/SPD 246, Anchorage, AK 2025

KRON4, NBC Bay Area, ABC7 Bay Area, and *San Francisco Chronicle* coverage of the 2024 Northern Lights 2024

## SELECTED PUBLICATIONS

**Mahajan, S. S.**, Upton, L. A., Antia, H. M., Basu, S., DeRosa, M. L., Hess Webber, S. A., et al.  
*The Sun's Large-Scale Flows I: Measurements of Differential Rotation & Torsional Oscillation.* Solar Physics 299, 38 2024

**Mahajan, S. S.**, Sun, X., and Zhao, J.  
*Removal of Active Region Inflows Reveals a Weak Solar Cycle Scale Trend in the Near-Surface Meridional Flow.* Astrophysical Journal 950, 63 2023

Dash, S., DeRosa, M. L., Dikpati, M., Sun, X., **Mahajan, S. S.**, Liu, Y., and Hoeksema, J. T.  
*Ensemble Kalman Filter Data Assimilation into the Surface Flux Transport Model to Infer Surface  
Flows.* Astrophysical Journal [2024](#)

## SKILLS

Python, MATLAB, Fortran, bash, OpenMP, MPI, C++, CUDA, scientific computing, time-series analysis, machine learning workflows, data analysis, visualization, and technical writing.

## EDUCATION

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|-----------------------------------------------------|----------------------|
| <i>Ph.D.</i> in Astronomy, Georgia State University | <a href="#">2019</a> |
| <i>M.S.</i> in Physics, Georgia State University    | 2017                 |
| <i>M.Tech.</i> in Engineering Physics, IIT (BHU)    | 2014                 |